

Answer one question from this section in the spaces provided.

- 8** Fig. 8.1 shows a thin iron strip of length 8.0 cm, width 2.0 cm and thickness 2.0 mm. As iron is a conductor of electricity, it contains free electrons, one of which is shown in Fig. 8.1.

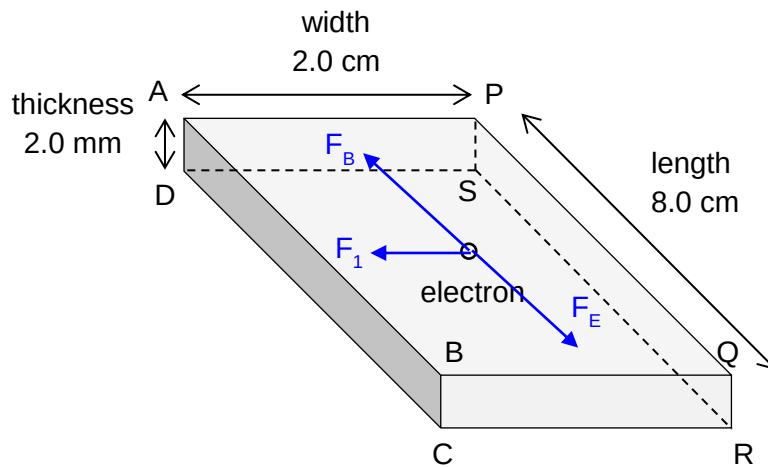


Fig. 8.1

- (a) A small potential difference is applied to the iron strip such that face ABCD is at a potential 12 mV higher than face PQRS.
- (i) Draw, on Fig. 8.1, an arrow to show the direction of the force on the free electron. Label as F_1 . [1]
- (ii) Calculate the magnitude of the force in (a)(i).

$$E = V / d = 12 \times 10^{-3} / 0.020 = 0.60 \text{ N C}^{-1}$$

$$F = q E = 1.6 \times 10^{-19} \times 0.60 = 9.6 \times 10^{-20} \text{ N}$$

force = N [2]

- (iii) Determine the change in the electric potential energy of the electron when it moves across the entire width of the strip.

$$\text{Loss in EPE} = q V = 1.6 \times 10^{-19} \times 12 \times 10^{-3} = 1.9 \times 10^{-21} \text{ J}$$

change in electric potential energy = J [2]

(b) Iron has a resistivity of $9.7 \times 10^{-8} \Omega \text{ m}$ and an electron density of $8.8 \times 10^{28} \text{ m}^{-3}$.

- (i) Calculate the current in the iron strip when there is a potential difference of 12 mV across faces ABCD and PQRS.

$$R = \rho L / A = 9.7 \times 10^{-8} \times 0.020 / (0.080 \times 0.0020) = 1.21 \times 10^{-5} \Omega$$

$$I = V / R = 12 \times 10^{-3} / 1.21 \times 10^{-5} = 990 \text{ A}$$

current = A [3]

- (ii) Hence, show that the drift velocity of the electrons is $4.4 \times 10^{-4} \text{ m s}^{-1}$.

$$I = n A v e$$

$$990 = 8.8 \times 10^{28} \times (0.080 \times 0.0020) \times v \times 1.6 \times 10^{-19}$$

$$v = 4.39 \times 10^{-4} \text{ m s}^{-1}$$

[1]

(c) A magnetic field of flux density 20 mT is now applied in the downward direction into face ABQP.

- (i) Explain what is meant by a magnitude of 20 mT.

A force per unit length of 20 mN m^{-1} is exerted by the magnetic field on a long conductor carrying 1 A of current placed perpendicular to the magnetic field.

..... [1]

- (ii) Draw, on Fig. 8.1, an arrow to show the direction of the force exerted by the magnetic field on the electron. Label as F_2 . [1]

- (iii) Calculate the magnitude of the force in (c)(ii).

$$F = B q v = 20 \times 10^{-3} \times 1.6 \times 10^{-19} \times 4.4 \times 10^{-4} = 1.41 \times 10^{-24} \text{ N}$$

force = N [2]

(iii) Explain why a potential difference develops across faces APSD and BQRC.

The magnetic force on electrons is parallel to PQ, causing them to drift in the direction of the force. Due to the resulting difference in number density of electrons across the two ends, an electric field is set up across the two faces. [1]

(iv) Draw, on Fig. 8.1, an arrow to show the direction of the force due to the potential difference in (c)(iii) on the electron. Label as F_3 . [1]

(v) The potential difference in (c)(iii) eventually reaches a steady value. State the magnitude of the resultant force on the electron that acts along the length of the strip.

resultant force = zero N [1]

(vi) Hence calculate the value of the potential difference in (c)(iii).

$$F_E = F_B$$

$$1.6 \times 10^{-19} \times V / 0.080 = 1.41 \times 10^{-24} \text{ N}$$

$$V = 7.04 \times 10^{-7} \text{ V}$$

potential difference = V [2]

(vii) Briefly explain how this iron strip can be used to determine the direction of a different magnetic field in another region.

Change orientation of strip until p.d. across APSD and BQRC is maximum.

(The magnetic field is normal to the face APQB.)

If APSD is at higher potential, the magnetic field is directed towards APQB. If APSD is at lower potential, the magnetic field is directed towards DSRC. [2]

[Total: 20]

- 9 (a) Radioactive decay is a random and spontaneous process